NEW STRONGLY REGULAR GRAPHS ON 1296 VERTICES FROM PARTIAL DIFFERENCE SETS IN $\mathbb{F}_{16} \times \mathbb{F}_{81}$

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Abstract. We construct strongly regular graphs with parameters $(1296, 560, 244, 240)$ and $(1296, 592, 268, 272)$, for which no construction was previously known. We also construct the first vertex-transitive graphs with parameters $(1296, 595, 274, 272)$, a parameter set realized independently by Kovaľ. All are Cayley graphs of $\mathbb{Z}_2^4 \times \mathbb{Z}_3^4$, realized as the additive group of $\mathbb{F}_{16} \times \mathbb{F}_{81}$, whose connection sets are unions of orbits of a small group of field multiplications and a Galois map. We find 10, 4 and 4 pairwise non-isomorphic graphs with the three parameter sets. The connection sets were found by an exact constraint-programming search for partial difference sets in products of finite fields, which extends Kovaľ's search in $\mathbb{F}_4 \times \mathbb{F}_{81}$ for $\mathrm{srg}(324, 136, 58, 56)$. The three families are linked by spreads of Delsarte cliques. Some of the graphs with the first parameter set arise from an $\mathrm{srg}(1296, 665, 340, 342)$ by deleting three pairwise orthogonal spreads; this $\mathrm{srg}(1296, 665, 340, 342)$ is the graph of a symmetric Bush-type Hadamard matrix, which answers a question of Kovaľ. The graphs with the third parameter set arise by deleting two. None of the graphs with Latin square parameters is the graph of an orthogonal array. In one of the graphs with the first parameter set, every coset of $\mathbb{F}_{81}$ induces the Brouwer–Haemers graph $\mathrm{srg}(81, 20, 1, 6)$.

1. Introduction

A strongly regular graph $\mathrm{srg}(v, k, \lambda, \mu)$ is a k-regular graph on v vertices, neither complete nor edgeless, in which any two adjacent vertices have exactly λ common neighbors and any two distinct non-adjacent vertices exactly μ. Brouwer maintains an online table of feasible parameter sets [2]; see also the monograph of Brouwer and Van Maldeghem [4]. All three parameter sets of Theorems 1.1 and 1.2 were marked there as open when this work began.

A standard route to strongly regular graphs is through partial difference sets (PDS). If D is a (v, k, λ, μ)-PDS in a group G with $D = D^{(-1)}$ and $1 \notin D$, then the Cayley graph $\mathrm{Cay}(G, D)$ is an $\mathrm{srg}(v, k, \lambda, \mu)$ [10, Proposition 1.1]. Brady [1] turned this into a systematic computational program aimed at *new* strongly regular graphs: a local search found PDSs with 62 parameter sets in 1254 non-isomorphic groups of order at most 147, including the first constructions of $\mathrm{srg}(144, 52, 16, 20)$ and $\mathrm{srg}(147, 66, 25, 33)$.

Recently Kovaľ [9, Section 5] realized the Latin square parameters $\mathrm{LS}_8(18) = (324, 136, 58, 56)$ and the negative Latin square parameters $\mathrm{NL}_6(18)$ and $\mathrm{NL}_7(18)$ by PDSs in $\mathbb{Z}_2^2 \times \mathbb{Z}_3^4$. One of these is a cyclotomic construction in the additive group of $\mathbb{F}_4 \times \mathbb{F}_{81}$: the connection set is invariant under multiplication by the fourth roots of unity $\mu_4 \leq \mathbb{F}_{81}^*$, and each of its four slices D_a, $a \in \mathbb{F}_4$, is a union of cyclotomic classes of index 20 in $\mathbb{F}_{81}$ [9, Construction 5.4].

This paper replaces $\mathbb{F}_4$ by $\mathbb{F}_{16}$ and searches systematically over the symmetry groups that the product structure provides (Section 3). On $1296 = 36^2$ vertices this produces the following.

Theorem 1.1. *There exist strongly regular graphs with parameters*

$$(1296, 560, 244, 240) \quad (\mathrm{LS}_{16}(36)) \qquad \textit{and} \qquad (1296, 592, 268, 272) \quad (\mathrm{NL}_{16}(36)).$$

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There are at least 10 *pairwise non-isomorphic graphs with the first parameter set and at least* 4 *with the second. All of them are vertex-transitive Cayley graphs of* $\mathbb{Z}_2^4 \times \mathbb{Z}_3^4$.

The parameters $(1296, 595, 274, 272) = \mathrm{LS}_{17}(36)$ were realized independently by Kovaľ [8], with six graphs, none of them vertex-transitive. Our search gives the first vertex-transitive examples.

Theorem 1.2. *There are at least* 4 *pairwise non-isomorphic vertex-transitive strongly regular graphs with parameters* $(1296, 595, 274, 272)$. *They are Cayley graphs of* $\mathbb{Z}_2^4 \times \mathbb{Z}_3^4$, *so their connection sets are abelian partial difference sets with these parameters, and none of them is isomorphic to a graph of [8].*

The three families are related by deleting spreads of Delsarte cliques (Section 5.3). Two of the graphs with parameters $\mathrm{LS}_{16}(36)$ arise from a Cayley graph with parameters $\mathrm{LS}_{19}(36)$ by deleting three pairwise orthogonal spreads, and this Cayley graph is the graph of a symmetric Bush-type Hadamard matrix. This answers a question of Kovaľ [8, Section 4.10] (Proposition 5.3). One of the two arises from an $\mathrm{LS}_{21}(36)$ Cayley graph by deleting five pairwise orthogonal spreads, the largest possible number in this group. The graphs of Theorem 1.2 arise from $\mathrm{LS}_{19}(36)$ Cayley graphs by deleting two orthogonal spreads, as Kovaľ's do.

2. Preliminaries

2.1. **Latin square type parameters.** The *Latin square* and *negative Latin square* parameters [10, Section 2] are, for $n \geq 2$ and $m \geq 1$,

$$\mathrm{LS}_m(n) = (n^2,\, m(n-1),\, n-2+(m-1)(m-2),\, m(m-1)),$$
$$\mathrm{NL}_m(n) = (n^2,\, m(n+1),\, -n+m^2+3m,\, m(m+1)).$$

For $m \leq n$ and $m \leq n-2$ respectively these are the parameters of graphs that are neither complete nor edgeless. An $\mathrm{LS}_m(n)$ graph has eigenvalues $n-m$ and $-m$; an $\mathrm{NL}_m(n)$ graph has eigenvalues m and $m-n$. The graph of an orthogonal array $\mathrm{OA}(n, m)$ (equivalently, of $m-2$ mutually orthogonal Latin squares of order n) is an $\mathrm{LS}_m(n)$ graph, and every edge of it lies in a clique of size n. For $n = 36$ only eight MOLS are known, so known orthogonal arrays give $\mathrm{LS}_m(36)$ graphs only for $m \leq 10$ (and their complements); Brouwer's table [2] lists $\mathrm{OA}(36, 10)$ as known and $\mathrm{OA}(36, 11)$ as open.

2.2. **Partial difference sets and characters.** Let G be a finite group of order v. A k-subset $D \subseteq G$ is a (v, k, λ, μ)-*partial difference set* (PDS) if the products gh^{-1} with $g, h \in D$, $g \neq h$, represent every non-identity element of D exactly λ times and every non-identity element outside D exactly μ times; it is *regular* if $1 \notin D$ and $D = D^{(-1)}$ [10, Section 1]. From now on G is abelian and written additively, so a regular PDS satisfies $0 \notin D = -D$. Let (v, k, λ, μ) be feasible with integral eigenvalues $r > s$, the roots of $x^2 - (\lambda - \mu)x - (k - \mu)$, and let $D \subseteq G \setminus \{0\}$ with $D = -D$ and $|D| = k$.

Lemma 2.1 ([10, Corollary 3.3])**.** *D is a (v, k, λ, μ)-PDS if and only if $\chi(D) = \sum_{d \in D} \chi(d) \in \{r, s\}$ for every non-principal character χ of G.*

Lemma 2.2 ([10, Theorem 4.1])**.** *Let D be a regular PDS in an abelian group G. If $(\lambda - \mu)^2 + 4(k - \mu)$ is a perfect square, then $tD = D$ for every integer t coprime to $|G|$.*

In particular, a PDS with integral eigenvalues is a union of orbits of the multiplier group $\{x \mapsto tx : \gcd(t, |G|) = 1\}$. Any additional group of automorphisms that we impose on D can therefore be taken to contain the multipliers, with no loss of generality.

3. The search

3.1. **Products of finite fields.** Let $G = \mathbb{F}_{q_1} \times \cdots \times \mathbb{F}_{q_\ell}$ (additive group) with $q_i = p_i^{e_i}$ and $v = \prod q_i$. Every $\boldsymbol{\gamma} = (\gamma_1, \ldots, \gamma_\ell) \in \prod \mathbb{F}_{q_i}^*$ acts on G by $(a_i) \mapsto (\gamma_i a_i)$, and every $\boldsymbol{\sigma} = (\sigma_1, \ldots, \sigma_\ell)$ with $\sigma_i \in \mathrm{Gal}(\mathbb{F}_{q_i}/\mathbb{F}_{p_i})$ acts by $(a_i) \mapsto (\sigma_i(a_i))$. Both are group automorphisms. For a subgroup $S \le \prod \mathbb{F}_{q_i}^*$ and an optional Galois element $\boldsymbol{\sigma}$ we let

$$K(S, \boldsymbol{\sigma}) = \langle\, S,\ \boldsymbol{\sigma},\ \{x \mapsto tx : \gcd(t, v) = 1\} \,\rangle \le \mathrm{Aut}(G).$$

A *model* is a pair $(S, \boldsymbol{\sigma})$. Its solutions are the $K(S, \boldsymbol{\sigma})$-invariant PDSs in G with prescribed parameters. The models $(S, \boldsymbol{\sigma})$ and $(S, \boldsymbol{\sigma}^{-1})$ give the same group. For $G = \mathbb{F}_{16} \times \mathbb{F}_{81}$ an integer t coprime to 6 acts as $(a, b) \mapsto (a, \pm b)$, so the multipliers contribute only $-1 = (1, \beta^{40})$.

3.2. **An exact formulation.** Let $O_1, \ldots, O_N$ be the orbits of $K = K(S, \boldsymbol{\sigma})$ on G, and let $\Omega_1, \ldots, \Omega_M$ be the orbits of the dual action of K on the character group $\widehat{G}$. The value $\chi(O_i)$ depends only on the dual orbit of χ. Because K contains all multipliers, $\chi(O_i)$ is fixed by $\mathrm{Gal}(\mathbb{Q}(\zeta_{\exp G})/\mathbb{Q})$, so $c_{ji} = \chi_j(O_i) \in \mathbb{Z}$ for a representative $\chi_j \in \Omega_j$. By Lemma 2.1, K-invariant (v, k, λ, μ)-PDSs correspond exactly to the 0/1 solutions (x, b) of

$$\sum_i |O_i|\, x_i = k, \qquad x_{\{0\}} = 0, \qquad \sum_j |\Omega_j|\, b_j = f,$$

$$\sum_i c_{ji}\, x_i = s + (r - s)\, b_j \quad (\Omega_j \neq \{1\}),$$

where f is the multiplicity of the eigenvalue r; the equation $\sum_j |\Omega_j|\, b_j = f$ is implied by the others but helps the solver. We solve this integer program with the CP-SAT solver of OR-Tools [16]. Every solution is checked independently by the full identity $A^2 = kI + \lambda A + \mu(J - I - A)$ for A the adjacency matrix of $\mathrm{Cay}(G, D)$.

3.3. **Relation to Koval's search.** Koval's cyclotomic search on $\mathbb{F}_4 \times \mathbb{F}_{81}$ [9, Section 5.4] is the analogue of this scheme for $(q_1, q_2) = (4, 81)$, except that it also allows non-cyclic Galois parts. There S runs over the 16 subgroups of even order of the cyclic group $\mathbb{F}_4^* \times \mathbb{F}_{81}^* \cong C_{240}$, and the Galois part over the 8 subgroups of $\mathrm{Gal}(\mathbb{F}_4/\mathbb{F}_2) \times \mathrm{Gal}(\mathbb{F}_{81}/\mathbb{F}_3)$. For $(q_1, q_2) = (16, 81)$ the group $\mathbb{F}_{16}^* \times \mathbb{F}_{81}^* \cong C_{15} \times C_{80}$ is not cyclic, and we let S run over all of its 80 subgroups. Here S need not have even order, because -1 is among the multipliers. $\boldsymbol{\sigma}$ runs over the trivial map and the 15 maps $(a, b) \mapsto (a^{2^u}, b^{3^w})$ with $(u, w) \neq (0, 0)$. This gives 1280 models. The 1251 models with at most 200 orbits were each tested against every parameter set $(1296, k, \lambda, \mu)$ with $k \le 648$ and integral eigenvalues that was open in Brouwer's table; complements cover the rest. Section 4 presents the solutions, which have the three parameter sets of Theorems 1.1 and 1.2. The same scheme was also run on other products of finite fields; the solver settings and the coverage of these searches are recorded in the data repository.

4. The graphs

Throughout, $\mathbb{F}_{16} = \mathbb{F}_2[x]/(x^4 + x^3 + 1)$ and $\mathbb{F}_{81} = \mathbb{F}_3[y]/(y^4 + y^3 + 2)$, with primitive elements $\alpha = x$ and $\beta = y$. Vertices are pairs $(a, b) \in \mathbb{F}_{16} \times \mathbb{F}_{81}$, and $(a, b) \sim (a', b')$ if and only if $(a - a', b - b') \in D$.

Construction 4.1 (Graph I)**.** Let

$$\begin{aligned} S_I = \{&(0,0), (0,2), (0,6), (0,11), (0,13), (0,14), (0,17), (0,18), (0,19),\\ &(1,3), (1,6), (1,7), (1,8), (1,10), (1,11), (1,13), (1,15), (1,19),\\ &(2,1), (2,2), (2,5), (2,9), (2,10), (2,11), (2,13), (2,16), (2,17)\} \subseteq \mathbb{Z}_3 \times \mathbb{Z}_{20}, \end{aligned}$$

and

$$D_I = \{(0,\beta^j) : j \equiv 0 \pmod 4\} \\ \cup\ \{(\alpha^i,\beta^j) : 0 \le i < 15,\ 0 \le j < 80,\ (i \bmod 3,\ (j - 4\lfloor i/3\rfloor) \bmod 20) \in S_I\}.$$

Then $|D_I| = 20 + 27 \cdot 20 = 560$. D_I is invariant under $(a,b) \mapsto (\alpha^3 a, \beta^4 b)$, which has order 20 and whose tenth power is -1, and under $(a,b) \mapsto (a^2, b^{27})$.

Construction 4.2 (Graph II)**.** Let $J_a \subseteq \mathbb{Z}_{20}$ for $a \in \mathbb{F}_{16}$ be given by Table 1, and

$$D_{II} = \{(a,\beta^j) : a \in \mathbb{F}_{16},\ 0 \le j < 80,\ j \bmod 20 \in J_a\}.$$

Then $|D_{II}| = 4\,(13 + 15 \cdot 9) = 592$. D_{II} is invariant under multiplication by $\mu_4 = \langle \beta^{20} \rangle$ on the second coordinate and under $(a,b) \mapsto (a^2, b^3)$. Each slice $D_a = \{b : (a,b) \in D_{II}\}$ is a union of the cyclotomic classes $\beta^j \mu_4$ of index 20. This is the shape of Koval's Construction 5.4 in [9], with $\mathbb{F}_4$ replaced by $\mathbb{F}_{16}$.

Construction 4.3 (Graph III)**.** Let $J'_a \subseteq \mathbb{Z}_{20}$ for $a \in \mathbb{F}_{16}$ be given by Table 1, and

$$D_{III} = \{(a,\beta^j) : a \in \mathbb{F}_{16},\ 0 \le j < 80,\ j \bmod 20 \in J'_a\}\ \cup\ (\mathbb{F}_4^* \times \{0\}),$$

where $\mathbb{F}_4^* = \{1, \alpha^5, \alpha^{10}\}$. Then $|D_{III}| = 4\,(7 + 3 \cdot 11 + 12 \cdot 9) + 3 = 595$. D_{III} is invariant under multiplication by μ_4 on the second coordinate and under $(a,b) \mapsto (a^2, b^3)$. The three slices with $|J'_a| = 11$ are those with $a \in \mathbb{F}_4^*$.

TABLE 1. The sets J_a of Construction 4.2 and J'_a of Construction 4.3.

a	J_a (Γ_{II})	J'_a (Γ_{III})
0	$\{0,1,3,4,7,8,9,11,12,13,16,17,19\}$	$\{0,4,5,8,12,15,16\}$
α^0	$\{0,2,6,11,13,14,17,18,19\}$	$\{0,2,5,6,11,13,14,15,17,18,19\}$
α^1	$\{4,6,7,10,11,12,13,15,19\}$	$\{5,9,10,12,13,14,16,18,19\}$
α^2	$\{1,5,10,12,13,16,17,18,19\}$	$\{2,7,8,10,14,15,16,17,19\}$
α^3	$\{1,2,7,9,10,14,15,18,19\}$	$\{3,4,5,6,10,12,13,14,19\}$
α^4	$\{3,8,10,11,14,15,16,17,19\}$	$\{1,2,4,5,6,8,10,11,17\}$
α^5	$\{0,1,5,6,9,13,14,15,17\}$	$\{0,2,3,4,6,7,11,14,16,18,19\}$
α^6	$\{1,2,3,5,6,7,10,14,17\}$	$\{2,9,10,12,15,16,17,18,19\}$
α^7	$\{0,1,5,6,7,8,10,13,14\}$	$\{0,2,4,5,9,10,11,16,17\}$
α^8	$\{2,4,5,8,9,10,11,13,17\}$	$\{3,4,6,10,11,12,13,15,18\}$
α^9	$\{3,5,6,7,9,10,13,14,18\}$	$\{1,2,4,8,10,11,13,15,18\}$
α^{10}	$\{0,2,3,5,7,11,15,18,19\}$	$\{0,1,2,6,8,9,12,13,14,17,18\}$
α^{11}	$\{0,2,7,9,10,11,15,16,18\}$	$\{0,3,8,10,12,14,15,17,19\}$
α^{12}	$\{1,2,3,6,9,10,11,15,18\}$	$\{5,6,7,8,10,11,14,16,17\}$
α^{13}	$\{0,3,5,6,9,10,12,14,17\}$	$\{0,1,4,5,10,13,16,18,19\}$
α^{14}	$\{0,1,2,3,4,10,15,18,19\}$	$\{0,6,7,8,10,11,12,13,15\}$

Theorem 4.4. *D_I, D_{II} and D_{III} are partial difference sets in $(\mathbb{F}_{16} \times \mathbb{F}_{81}, +) \cong \mathbb{Z}_2^4 \times \mathbb{Z}_3^4$ with parameters $(1296, 560, 244, 240)$, $(1296, 592, 268, 272)$ and $(1296, 595, 274, 272)$. Consequently the Cayley graphs Γ_I, Γ_{II}, Γ_{III} of $\mathbb{F}_{16} \times \mathbb{F}_{81}$ with these connection sets are strongly regular, with spectra $560^1\,20^{560}\,(-16)^{735}$, $592^1\,16^{703}\,(-20)^{592}$ and $595^1\,19^{595}\,(-17)^{700}$.*

Proof. All three sets satisfy $D = -D$ and $0 \notin D$. The identity $A^2 = kI + \lambda A + \mu(J - I - A)$ was verified exhaustively, by counting the common neighbors of all $\binom{1296}{2}$ vertex pairs of $\mathrm{Cay}(G, D)$ (Section 6). As a consistency check, all 1295 non-principal character values were computed

TABLE 2. Non-isomorphic graphs found, with $|\mathrm{Aut}\Gamma|$ computed by nauty, the order of the multiplicative stabilizer $\mathrm{Stab}(D) = \{\gamma \in \mathbb{F}_{16}^* \times \mathbb{F}_{81}^* : \gamma D = D\}$ of the connection set, the clique number ω, and the model in which each graph was first found. A multiplicative generator (α^x, β^y) acts by $(a, b) \mapsto (\alpha^x a, \beta^y b)$, and a Galois map is written by its action on (a, b).

graph	parameters	$\lvert\mathrm{Aut}\Gamma\rvert$	$\lvert\mathrm{Stab}(D)\rvert$	ω	model $(S, \boldsymbol{\sigma})$
I.1	$(1296, 560, 244, 240)$	103680	20	12	$\langle(\alpha^3, \beta^4)\rangle$, (a^2, b^{27})
I.2		51840	20	12	$\langle(\alpha^3, \beta^{12})\rangle$, (a^4, b^9)
I.3		51840	10	12	$\langle(\alpha^3, \beta^{16})\rangle$, (a^2, b^{27})
I.4		51840	20	12	$\langle(\alpha^3, \beta^{12})\rangle$, none
I.5		25920	20	12	$\langle(\alpha^3, \beta^{36})\rangle$, none
I.6		25920	20	12	$\langle(\alpha^3, \beta^4)\rangle$, none
I.7		20736	4	12	$\langle(1, \beta^{20})\rangle$, (a^2, b^3)
I.8		20736	4	15	$\langle(1, \beta^{20})\rangle$, (a^2, b^{27})
I.9		20736	4	12	$\langle(1, \beta^{20})\rangle$, (a^8, b^3)
I.10		20736	4	14	$\langle(1, \beta^{20})\rangle$, (a^8, b^{27})
II.1	$(1296, 592, 268, 272)$	20736	4	13	$\langle(1, \beta^{20})\rangle$, (a^2, b^3)
II.2		20736	4	12	$\langle(1, \beta^{20})\rangle$, (a^2, b^{27})
II.3		20736	4	12	$\langle(1, \beta^{20})\rangle$, (a^8, b^3)
II.4		20736	4	12	$\langle(1, \beta^{20})\rangle$, (a^8, b^{27})
III.1	$(1296, 595, 274, 272)$	20736	4	24	$\langle(1, \beta^{20})\rangle$, (a^2, b^3)
III.2		20736	4	24	$\langle(1, \beta^{20})\rangle$, (a^2, b^{27})
III.3		20736	4	24	$\langle(1, \beta^{20})\rangle$, (a^8, b^3)
III.4		20736	4	24	$\langle(1, \beta^{20})\rangle$, (a^8, b^{27})

numerically. They take exactly the two values r, s allowed by Lemma 2.1, with the multiplicities of the stated spectrum. □

Remark 4.5. Unlike Koval's Construction 5.4, which is proved by hand in [9, Section 5.3], these proofs are computational. A proof in that style should be possible, since every character value of D_I, D_{II} and D_{III} is a sum of Gauss periods of index 20 of $\mathbb{F}_{81}$ with signs ± 1 given by additive characters of $\mathbb{F}_{16}$.

4.1. **Further graphs.** Table 2 lists all isomorphism classes found. Graphs I.7–I.10, II.1–II.4 and III.1–III.4 come from the same four models $(S, \boldsymbol{\sigma})$, with $S = \langle(1, \beta^{20})\rangle \cong \mu_4$ and one of four Galois maps. Since (a^8, b^{27}) and (a^8, b^3) are the inverses of (a^2, b^3) and (a^2, b^{27}), these four models define only two groups K, each of which has solutions for all three parameter sets. Of Koval's six graphs with parameters $(1296, 595, 274, 272)$ [8], four have automorphism groups of orders 2903040, 3456, 31104 and 19584 [8, Table 4], and nauty [11] gives trivial automorphism groups for the other two. None of them is vertex-transitive, so none is isomorphic to III.1–III.4, which proves the last claim of Theorem 1.2; a comparison of nauty canonical forms confirms this. The connection sets of all eighteen graphs are in the data repository (see Data availability).

Graphs I.1, II.1 and III.1 are Γ_I, Γ_{II} and Γ_{III} of Constructions 4.1–4.3. All eighteen graphs are vertex-transitive, being Cayley graphs. The counts are lower bounds, since the search is not exhaustive.

5. PROPERTIES

5.1. **Clique numbers.** In the graph of an $\mathrm{OA}(36, m)$ every edge lies in a 36-clique, which attains the Delsarte bound $1 - k/s = 36$ on the size of a clique [4]. We computed the clique

numbers ω of all eighteen graphs exactly (Table 2). Since Γ is vertex-transitive, a maximum clique may be taken to contain $(0,0)$ and an edge $\{(0,0),x\}$, so $\omega(\Gamma)=2+\max_x \omega(\Gamma(0,0)\cap\Gamma(x))$. Here x runs over representatives of the orbits of the known automorphisms fixing $(0,0)$ on its neighbors. Each common neighborhood has λ vertices, and its clique number was computed exactly with Cliquer [13] (through igraph). A clique of size ω was exhibited in each case, and the values for I.1 and I.8 were confirmed with a second exact solver (networkx). The clique numbers are 12, 14 or 15 for the graphs I, 12 or 13 for the graphs II and 24 for the graphs III. They are far below the Delsarte bounds, which are 36 for I and III and $\lfloor 1+592/20\rfloor = 30$ for II. In particular none of the graphs I and III is the graph of an orthogonal array, and they say nothing about the existence of 14 or 15 MOLS of order 36.

5.2. **Ring structure.** Let $R_a=\{a\}\times\mathbb{F}_{81}$ for $a\in\mathbb{F}_{16}$; we call these sets *rings*. Since they are the cosets of a subgroup, $\{R_a\}$ is an equitable partition of every Cayley graph of G.

Proposition 5.1. *In Γ_I each ring induces the Brouwer–Haemers graph* $\mathrm{srg}(81,20,1,6)$ *[3], namely the cyclotomic graph* $\mathrm{Cay}(\mathbb{F}_{81},\langle\beta^4\rangle)$*, and every vertex has exactly* 36 *neighbors in each of the other* 15 *rings. In Γ_{II} each ring induces a* 52*-regular graph that is not strongly regular, and every vertex has exactly* 36 *neighbors in each other ring. In Γ_{III} each ring induces a* 28*-regular graph that is not strongly regular. A vertex of R_a has* 45 *neighbors in each of the three rings R_{a+c}, $c\in\mathbb{F}_4^*$, and* 36 *in each of the other twelve rings.*

Proof. The subgraph induced on R_a is $\mathrm{Cay}(\mathbb{F}_{81}, D\cap(\{0\}\times\mathbb{F}_{81}))$, and $D_I\cap(\{0\}\times\mathbb{F}_{81})=\{0\}\times\langle\beta^4\rangle$. This cyclotomic graph is an $\mathrm{srg}(81,20,1,6)$, as a direct count shows, and an $\mathrm{srg}(81,20,1,6)$ is unique [3], so it is the Brouwer–Haemers graph. The remaining counts are $|D\cap(\{a\}\times\mathbb{F}_{81})|$ for $a\neq 0$, read off from the constructions. For Γ_{II}, adjacent pairs in a ring have 31 or 37 common neighbors inside the ring, and for Γ_{III} they have 1, 7 or 13. □

The quotient matrix of the partition $\{R_a\}$ is $cI+36(J-I)$ with $c=20$ for Γ_I and $c=52$ for Γ_{II}. Its eigenvalues are $c+15\cdot 36=k$ and $c-36$, which equals -16 for Γ_I and 16 for Γ_{II}, as they must be. For Γ_{III} the quotient matrix is not of this form, because the subfield $\mathbb{F}_4$ singles out three rings, and it has eigenvalues 595, 19 and -17. Figure 1 illustrates the ring structure, and its right-hand panels zoom into a single ring. An interactive visualization of all eighteen graphs, including their adjacency matrices, is in the data repository. The ring graphs of Γ_{II} show that strong regularity of Γ does not require strongly regular rings: two non-adjacent vertices of the same ring have 26, 32 or 38 common neighbors inside the ring (adjacent ones 31 or 37), and the other 15 rings contribute exactly the complementary amount.

5.3. **Spread structure.** Following Haemers and Tonchev [6], a *spread* of a strongly regular graph is a partition of its vertex set into Delsarte cliques, that is, cliques of size $1-k/s$. Two spreads are *orthogonal* if every clique of one meets every clique of the other in exactly one vertex. Kovaľ [8, Lemma 4.1], building on [6, Proposition 4.1], showed the following: if $\mathcal{S}_1,\dots,\mathcal{S}_j$ are pairwise orthogonal spreads of a primitive strongly regular graph with the parameters of $\mathrm{LS}_m(n)$, $j<m$, then deleting the edges inside all their cliques leaves a strongly regular graph with the parameters of $\mathrm{LS}_{m-j}(n)$. Kovaľ constructed $\mathrm{LS}_{n/2-1}(n)$ graphs this way, by deleting two orthogonal spreads from $\mathrm{LS}_{n/2+1}(n)$ graphs coming from Bush-type Hadamard matrices, and asked whether a third orthogonal spread can be deleted, which for $n=36$ would give the parameters $\mathrm{LS}_{16}(36)=(1296,560,244,240)$ of Γ_I [8, Section 4.10].

In a Cayley graph $\mathrm{Cay}(G,E)$, a subgroup U of order 36 with $U\setminus\{0\}\subseteq E$ makes the cosets of U cliques of size 36. For an $\mathrm{LS}_m(36)$ graph these are Delsarte cliques, since $1-k/s=1+35m/m=36$. So the cosets of U form a spread, and the spreads of two such subgroups U,V are orthogonal exactly when $U\cap V=\{0\}$. Deleting the spread of U replaces E by $E\setminus U$. Conversely, we call

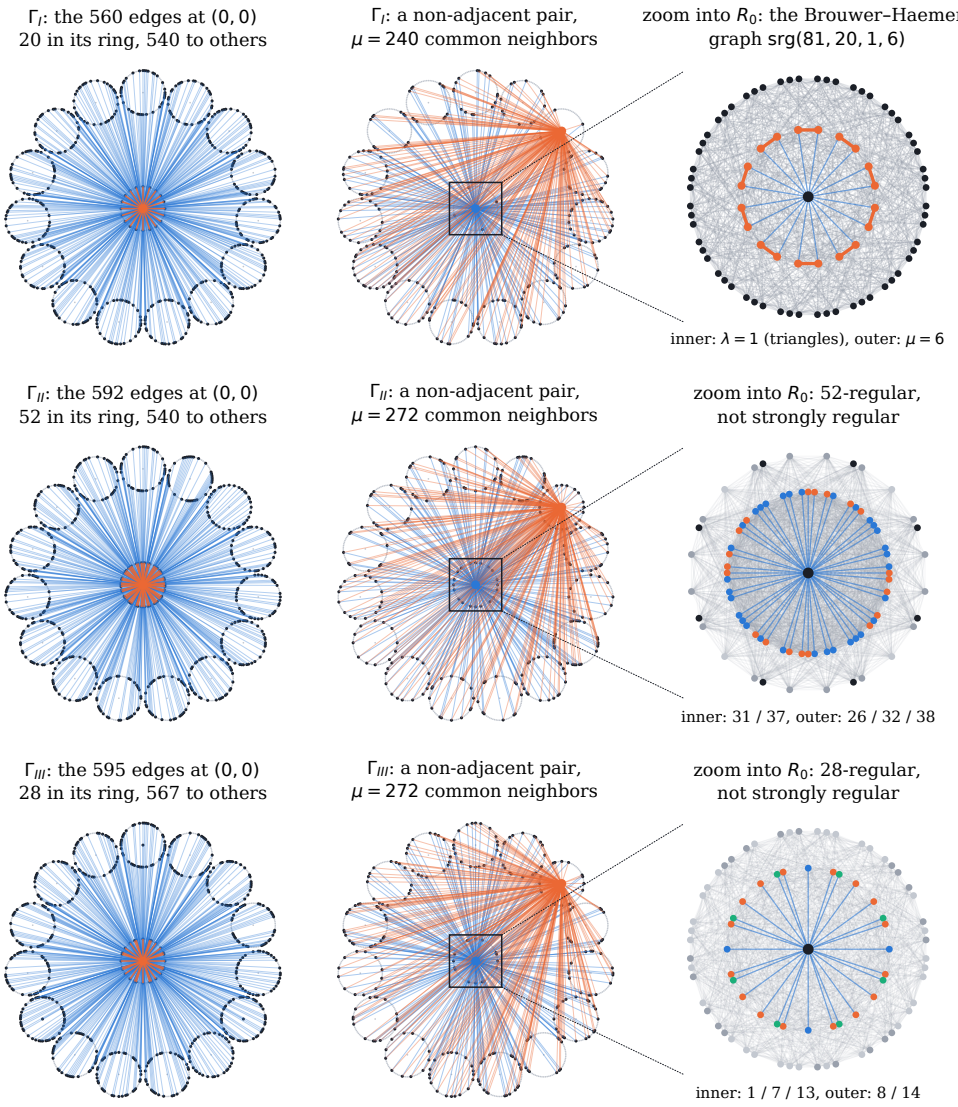
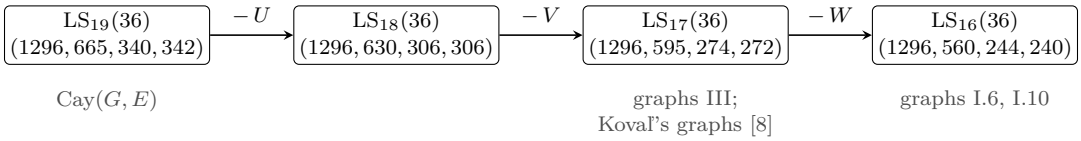
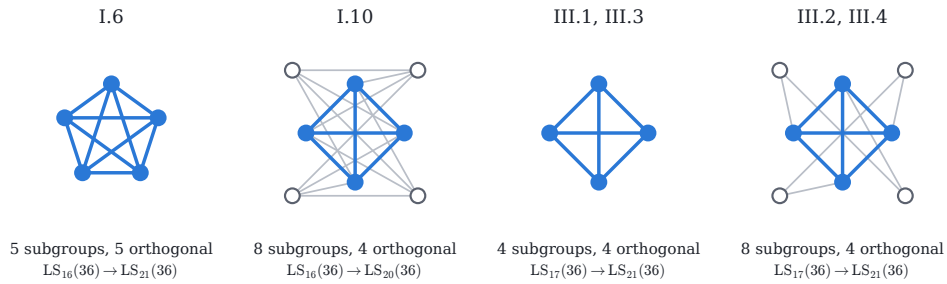


FIGURE 1. Rows: Γ_I, Γ_{II}, Γ_{III}. Left and middle: all 1296 vertices in the ring layout. Ring R_{α^i} is centered at angle $2\pi i/15$ and R_0 is in the middle; within a ring, (a, β^j) sits at angle $2\pi j/80$ with $(a, 0)$ at its center. Left: the edges at $(0,0)$, orange inside its ring and blue to other rings. Middle: a non-adjacent pair, $(0,0)$ (blue) and a vertex of $R_{\alpha^{13}}$ (orange), with the edges from both to their μ common neighbors (black dots). Right: the subgraph induced on the framed ring R_0. Here $(0,0)$ is in the center, its neighbors in the ring on the inner circle and its non-neighbors in the ring on the outer circle. Each vertex is colored by its number of common neighbors with $(0,0)$ inside the ring. In Γ_I these numbers are constant, $\lambda = 1$ and $\mu = 6$: the ring is the Brouwer–Haemers graph $\mathrm{srg}(81, 20, 1, 6)$, and the 20 neighbors of $(0,0)$ form ten triangles with it. In Γ_{II} and Γ_{III} the ring graphs are 52- and 28-regular but not strongly regular. The inner numbers are 31 or 37 (blue, orange) for Γ_{II} and 1, 7 or 13 (blue, orange, green) for Γ_{III}. The outer numbers are 26, 32 or 38 and 8 or 14 (light to dark).



(a)



(b)

FIGURE 2. (a) Deleting pairwise orthogonal spreads (Propositions 5.2 and 5.3). Each arrow replaces the connection set E by $E \setminus U$ for a subgroup U of order 36, and U, V, W meet pairwise in $\{0\}$. In each case $\mathrm{Cay}(G, E)$ is the graph Γ_0 of a symmetric Bush-type Hadamard matrix. The graphs III and I.6, I.10 come from different sets E. (b) Free subgroups of order 36 (Proposition 5.2). Vertices are the subgroups of order 36 disjoint from D, and two of them are joined if they intersect in $\{0\}$, that is, if their spreads are orthogonal. The largest set of pairwise orthogonal ones is shown in blue; adding all of them to D gives the Latin square type PDS shown below each graph. The other graphs I and all graphs II have no free subgroups.

a subgroup U of order 36 *free* for D if $U \cap D = \emptyset$; adding $U^* = U \setminus \{0\}$ to D may or may not give a PDS.

Proposition 5.2. *Free subgroups exist only for the graphs I.6, I.10 and III.1–III.4, whose connection sets D are of type $\mathrm{LS}_m(36)$ with $m = 16$ and $m = 17$. For these graphs, if $U_1, \ldots, U_j$ are free subgroups with pairwise intersection $\{0\}$, then $D \cup U_1^* \cup \cdots \cup U_j^*$ is an $\mathrm{LS}_{m+j}(36)$ PDS. The free subgroups are as follows (Figure 2(b)).*

(1) *I.6 has 5 free subgroups, and they intersect pairwise in $\{0\}$. So $D_{\mathrm{I.6}} \cup U_1^* \cup \cdots \cup U_5^*$ is an $\mathrm{LS}_{21}(36)$ PDS, and $\Gamma_{\mathrm{I.6}}$ arises from an $\mathrm{LS}_{21}(36)$ Cayley graph by deleting five pairwise orthogonal spreads. By the theorem of Bailey and Jungnickel (Section 5.4), five is the largest possible number of subgroups of order 36 of G that intersect pairwise in $\{0\}$.*
(2) *I.10 has 8 free subgroups, with 18 pairs, 8 triples and one set of four that intersect pairwise in $\{0\}$. So $\Gamma_{\mathrm{I.10}}$ arises from an $\mathrm{LS}_{20}(36)$ Cayley graph by deleting four pairwise orthogonal spreads.*
(3) *III.1 and III.3 have 4 free subgroups each, and they intersect pairwise in $\{0\}$. III.2 and III.4 have 8 each, with 12 pairs, 4 triples and one set of four that intersect pairwise in $\{0\}$. So every graph III arises from an $\mathrm{LS}_{21}(36)$ Cayley graph by deleting four pairwise orthogonal spreads.*

In particular $\Gamma_{\mathrm{I.6}}$ and $\Gamma_{\mathrm{I.10}}$ arise from an $\mathrm{LS}_{19}(36)$ Cayley graph by deleting three pairwise orthogonal spreads, in 10 and 8 ways, and every graph III arises from an $\mathrm{LS}_{19}(36)$ Cayley graph by deleting two.

Proof. By exhaustive computation over the $35 \cdot 130$ subgroups of order 36, which are the products of a subgroup of order 4 of $\mathbb{F}_{16}$ (there are 35) and a subgroup of order 9 of $\mathbb{F}_{81}$ (there are 130). Every set of free subgroups with pairwise trivial intersection was tested with the character criterion of Lemma 2.1. For one triple of each of I.6 and I.10, strong regularity of all four Cayley graphs $\mathrm{Cay}(G, D)$, $\mathrm{Cay}(G, D \cup U^*)$, $\mathrm{Cay}(G, D \cup U^* \cup V^*)$ and $\mathrm{Cay}(G, D \cup U^* \cup V^* \cup W^*)$ was also checked by counting common neighbors. □

Kovaľ [8, Theorem 4.3] obtains $\mathrm{LS}_{n/2+1}(n)$ graphs Γ_0 from symmetric Hadamard matrices K of order n^2 of Bush type whose blocks have unit diagonal: rows and columns are indexed by $[n] \times [n]$, the diagonal blocks are J_n, the other blocks K_{ac} satisfy $K_{ac}J_n = J_nK_{ac} = 0$, and $K_{(a,b),(c,b)} = 1$. The block rows and the block columns of Γ_0 are two orthogonal spreads, and deleting both gives an $\mathrm{LS}_{n/2-1}(n)$ graph Γ_K. He asked whether some Γ_0 has a third spread orthogonal to both [8, Section 4.10]. The Cayley graphs of Proposition 5.2 answer this question positively for $n = 36$.

Proposition 5.3. *Let E be an $\mathrm{LS}_{19}(36)$ PDS in G, and let U, V be subgroups of order 36 with $U \cap V = \{0\}$ and $U^* \cup V^* \subseteq E$. Index the vertices $g \in G$ by the pair $(g+U, g+V)$ of cosets, and let $K = 2(A + I) - J$, where A is the adjacency matrix of $\mathrm{Cay}(G, E)$. Then K is a symmetric Hadamard matrix of Bush type whose blocks have unit diagonal, $\mathrm{Cay}(G, E)$ is its graph Γ_0, and $\mathrm{Cay}(G, E \setminus (U \cup V))$ is its graph Γ_K. If moreover W is a subgroup of order 36 with $W^* \subseteq E$ and $W \cap U = W \cap V = \{0\}$, then the cosets of W form a third spread of Γ_0, orthogonal to the block rows and the block columns.*

Proof. Since $G = U \oplus V$, each U-coset meets each V-coset in exactly one vertex, so the indexing is a bijection. $\mathrm{Cay}(G, E)$ is an $\mathrm{srg}(1296, 665, 340, 342)$ with eigenvalues 665, 17 and -19, so K is a symmetric ± 1 matrix with eigenvalues 36 and ± 36, and $K^2 = 1296I$. The diagonal blocks are J because the U-cosets are cliques, and $K_{(a,b),(c,b)} = 1$ because the V-cosets are cliques. A vertex outside a Delsarte clique of size 36 has exactly $\mu/(-s) = 342/19 = 18$ neighbors in it [4], so every row of an off-diagonal block has 18 entries 1 and 18 entries -1; hence $K_{ac}J = 0$, and $JK_{ac} = 0$ by symmetry. The adjacency matrix of $\mathrm{Cay}(G, E)$ is $\frac{1}{2}(J + K) - I$, which is Γ_0, and deleting the spreads of U and V gives Γ_K. Finally, $W \cap U = W \cap V = \{0\}$ means that every W-coset meets every block row and every block column in one vertex, and $W^* \subseteq E$ makes the W-cosets cliques. □

By Proposition 5.2, Proposition 5.3 applies to every graph III, with $E = D \cup U^* \cup V^*$ for a pair of orthogonal free subgroups, so the graphs III are graphs Γ_K of Bush-type Hadamard matrices developed over G, like Kovaľ's $\mathrm{LS}_{17}(36)$ graphs. For I.6 and I.10, with $E = D \cup U^* \cup V^* \cup W^*$, it also gives the third spread. All Bush-type and spread statements above were also checked directly on the matrices.

5.4. **Relation to known constructions.** We checked the constructions of Latin square and negative Latin square type PDSs that could apply to abelian groups of order $1296 = 2^4 \cdot 3^4$. None of the constructions we checked gives any of our three parameter sets.

- *Partial congruence partitions.* In an abelian group of order 36^2, a set of subgroups of order 36 that intersect pairwise trivially has at most $\min(2^2 + 1, 3^2 + 1) = 5$ members, by a theorem of Bailey and Jungnickel [10, Theorem 2.4]. So these give $\mathrm{LS}_m(36)$ only for $m \leq 5$.
- *Product constructions.* The product theorems for packings of Latin square and negative Latin square type PDSs [7, Theorems 3.15, 6.8, Corollaries 3.17, 6.10], including Polhill's product construction [17], combine packings in groups G_1, G_2 of orders n_1^2, n_2^2 with $n_1n_2 = n$. They give $\mathrm{LS}_m(n)$ or $\mathrm{NL}_m(n)$ only when m is a multiple of n/t with $t > 1$

dividing $\gcd(n_1, n_2)$. For $n = 36$ we have $\gcd(n_1, n_2) \le 6$, so m is a multiple of 6. Our values $m = 16$ (LS and NL) and $m = 17$ (LS) are not, and neither are the complementary values 21, 19 and 20.

- *Momihara's families* [12, Theorems 1.4, 1.5] live in products of elementary abelian groups of orders q_i^4 with $2m \mid q_i + 1$ for all i, for an integer m (not the Latin square index). For $1296 = 2^4 \cdot 3^4$ this would require $2m \mid 3$ and $2m \mid 4$, which is impossible.
- *Cyclotomy over products of finite fields.* The standard cyclotomy of Fernández-Alcober, Kwashira and Martínez [5] on $\mathbb{F}_{16} \times \mathbb{F}_{81}$ has order $e \mid \gcd(15, 80) = 5$. Its PDSs are unions of orbits of a subgroup $H \le \mathbb{F}_{16}^* \times \mathbb{F}_{81}^*$ of index e. For $e = 1$ the orbits have sizes $1, 15, 80, 1200$, so no union has size 560, 592 or 595. For $e = 5$ there are four choices of H, the orbits have sizes 1, 15, 80 and 240, and an exhaustive check shows that none of the 40 unions of H-orbits of size 560 is a PDS; no union of H-orbits has size 592 or 595. So this cyclotomy gives none of our parameter sets. Independently, our connection sets are not of this form: their multiplicative stabilizers have order at most 20 (Table 2), whereas a union of H-orbits is invariant under H, which has order $1200/e \ge 240$. Ott's generalizations [14, 15] consider unions of cosets of $U_\chi = \{(x_1, x_2) : \chi_1(x_1)\chi_2(x_2) = 1\}$, together with U_χ-orbits of non-units, for multiplicative characters χ_i of order $d \mid q_i - 1$. As Ott notes [15, Section 1], these are exactly the subgroups of [5]. So here $d \mid 5$, and the same two arguments exclude them. The new family of [14, Theorem 3] lives in groups of prime-power order q^{2a+1}.

6. Verification

Each claimed graph was checked in three independent ways, and all checks pass.

(1) A standalone program (plain Python, no external packages, no code shared with the search) builds $\mathrm{Cay}(G, D)$ with its own arithmetic in $\mathbb{F}_{16}$ and $\mathbb{F}_{81}$. It checks $D = -D$, $0 \notin D$, the degree, and the number of common neighbors of all 839,160 vertex pairs. It is run on Constructions 4.1–4.3 exactly as printed above, and on the connection sets of all eighteen graphs.
(2) Koval's independent verifier `verify_srg.py`, from the data accompanying [9], was run on the graph6 files of all eighteen graphs.
(3) The spectra were computed numerically, and isomorphism classes and automorphism groups were computed with nauty [11].

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Use of AI systems. The search programs, the verification programs, the figures and a first draft of this text were produced in collaboration with an AI assistant (Claude Opus 5.5, Anthropic). All results were checked as described in Section 6. The author has verified every proof and every computational claim, and has written the final text in their own words.

Data availability. The connection sets of all graphs, the graphs in graph6 format, the standalone builders and verifiers, an interactive visualization, and the search code are available at `https://github.com/dylan-plummer/srg1296`.

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